PML Assignment 2: Self-Supervised Denoising for Noise-Level Selection

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1 Overview

We compare two noise-level (σ) selection strategies for self-supervised denoising on the **Iris** dataset (n=150, d=4, K=3): (1) σ_{proxy} – chosen *without labels* by minimising reconstruction MSE; (2) σ_{down} – chosen *with few labels* (m=5/class) by maximising test accuracy. Features are standardised using *training-split statistics only* to prevent leakage. Ridge regression (α=1) is the denoiser; multinomial logistic regression is the classifier. In Part B the pretrained denoiser g_σ is used as a *consistent feature transformer*: clean inputs are passed through g_σ for both labeled *and* test sets, avoiding any train/test mismatch.

2 Results

2.1 Part A – Proxy-Task (No Labels)

Features are split 75/25 (seed 42). For each σ, noise is added, a Ridge denoiser is fitted, and MSE is evaluated on the held-out set.

Table 1: σ vs. Proxy-Val MSE

σ	Proxy-Val MSE
0.0	0.000685
0.1	0.008499
0.2	0.027897
0.3	0.054187
0.4	0.085045
0.6	0.155183
0.8	0.230314

Rule: smallest MSE among $\sigma > 0$.
Result: $\boldsymbol{\sigma_{\text{proxy}} = 0.1}$.
σ=0 is excluded (trivial identity). Among $\sigma > 0$, the lowest noise yields the best reconstruction.

2.2 Part B – Downstream (Few Labels)

Data split 50/50 (stratified, seed 42). m=5 labeled per class. Clean features are passed through g_σ for both labeled training and test sets.

Table 2: σ vs. Test Accuracy (m=5/class)

σ	Test Accuracy
0.0	0.8533
0.1	0.8533
0.2	0.8533
0.3	0.8533
0.4	0.8667
0.6	0.8667
0.8	0.8667

Rule: highest test accuracy; ties broken by smallest σ.
Result: $\boldsymbol{\sigma_{\text{down}} = 0.4}$.

2.3 Multi-Seed Variance (Part B, 10 seeds)

The single-seed results above show many ties (accuracy differences ≤ 0.013). To assess stability, Part B is repeated over 10 independently sampled labeled sets.

Table 3: σ vs. Mean $\pm$ Std Test Accuracy (10 seeds)

σ	Mean Accuracy	Std
0.0	0.8893	0.0387
0.1	0.8867	0.0418
0.2	0.8827	0.0473
0.3	0.8787	0.0411
0.4	0.8693	0.0453
0.6	0.8560	0.0347
0.8	0.8573	0.0343

Averaging over seeds reveals that σ=0 achieves the highest mean accuracy (0.8893). The single-seed σ_{down}=0.4 lies within one standard deviation of the best, confirming that the slight advantage of higher σ in the single-seed run is due to sampling variability in the small labeled budget.

3 Discussion

3.1 Why Do the Two Strategies Disagree?

The proxy task optimises *reconstruction fidelity*, while the downstream task optimises *classification accuracy* – fundamentally different objectives. σ_{proxy}=0.1 minimises reconstruction error among non-trivial noise levels: small noise forces the denoiser to learn data structure while keeping distortion low. σ_{down}=0.4 (single seed) arises because the denoiser, as a feature transformer, applies a mild low-pass filter that can help with certain labeled subsets. However, the multi-seed analysis shows σ=0 is best on average, indicating that on Iris the raw features are already informative enough that denoising adds little benefit. This divergence illustrates that proxy and downstream objectives can favour different σ values.

3.2 Is There a "Sweet Spot"?

For **reconstruction**, MSE increases monotonically with σ; the sweet spot is the smallest positive σ (0.1). For **classification**, the multi-seed means decrease gradually from σ=0 (0.8893) to σ=0.6 (0.8560). The differences are small relative to the standard deviations ($\approx$0.04), meaning the "sweet spot" is broad and centered at σ=0. On higher-dimensional or noisier datasets, one would expect a clear interior peak where moderate denoising regularises representations and improves generalisation.

3.3 Proxy-Based vs. Downstream Tuning

Proxy-based (σ_{proxy}): requires *no labels*; scalable and stable. However, the proxy objective may not align with downstream needs.

Downstream (σ_{down}): directly optimises the target metric, but requires labeled data and is more sensitive to sampling variance with small budgets (as the multi-seed analysis demonstrates).

When to prefer each: use proxy tuning when labels are unavailable; use downstream tuning when even a small labeled set exists, supplemented by multi-seed averaging to reduce variance.

3.4 Reproducibility

Three sources of randomness are fixed: (1) `random_state=42` in `train_test_split` ensures identical partitions; (2) `RandomState(42)` for labeled-set sampling guarantees the same m=5 examples per class; (3) distinct seed offsets (`SEED+1`, `+2`) provide independent noise realisations while remaining reproducible. Parts A and B use separate train/test splits (75/25 vs. 50/50) since they are independent experiments; within each part, splits and labeled budgets are held constant across all σ values.